Haciendo uso de mongosh, escribe los comandos necesarios y adjunta el resultado de las consultas sobre la colección trips de la base de datos sample_training:

1. Recupera los viajes realizados con la bicicleta con ID 14785

```
> db.trips.find({bikeid: 14785})
< {
    _id: ObjectId('572bb8222b288919b68abf82'),
    tripduration: 812,
    'start station id': 518,
    'start station name': 'E 39 St & 2 Ave',
    'end station id': 433,
    'end station name': 'E 13 St & Avenue A',
    bikeid: 14785,
    usertype: 'Subscriber',
    'birth year': 1977,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.9734419,
        40.74780373
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -73.98057249,
        40.72955361
      ]
    },
    'start time': 2016-01-01T00:14:48.000Z,
    'stop time': 2016-01-01T00:28:21.000Z
  }
  {
```

2. Encuentra los viajes que duraron más de 1 hora.



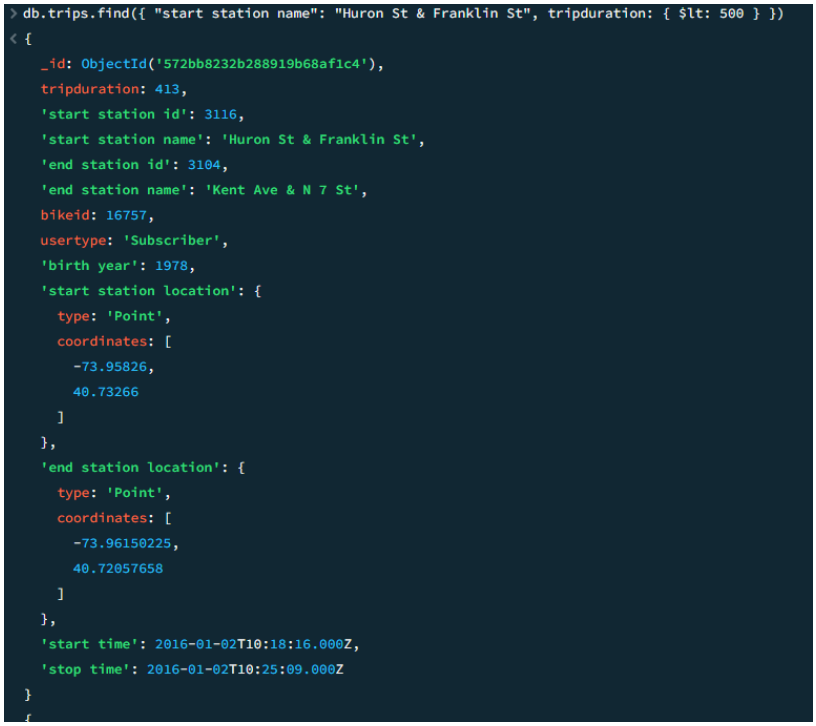
```
> db.trips.find({tripduration: {$gt: 3600}})
< {
    _id: ObjectId('572bb8222b288919b68ac07c'),
    tripduration: 326222,
    'start station id': 391,
    'start station name': 'Clark St & Henry St',
    'end station id': 310,
    'end station name': 'State St & Smith St',
    bikeid: 18591,
    usertype: 'Subscriber',
    'birth year': 1979,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.99344559,
        40.69760127
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -73.98912867,
        40.68926942
      ]
    },
    'start time': 2016-01-01T00:58:20.000Z,
    'stop time': 2016-01-04T19:35:23.000Z
  }
```

3. Mostrar solo el ID de la bicicleta, la duración del viaje y los nombres de las estaciones de los viajes realizados por usuarios nacidos después de 1990.



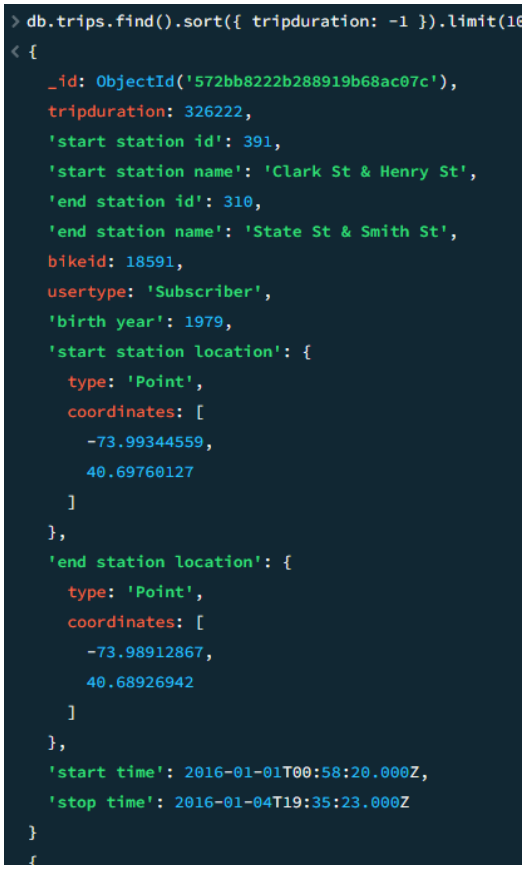
```
> db.trips.find({ "birth year": { $gt: 1990 } }, { bikeid: 1, tripduration: 1, "start station name": 1, "end station name": 1 })
< {
    _id: ObjectId('572bb8222b288919b68abf60'),
    tripduration: 694,
    'start station name': 'Howard St & Centre St',
    'end station name': 'E 17 St & Broadway',
    bikeid: 15747
  }
  {
    _id: ObjectId('572bb8222b288919b68abf6a'),
    tripduration: 479,
    'start station name': 'Franklin St & Dupont St',
    'end station name': 'Kent Ave & N 7 St',
    bikeid: 23563
  }
```

4. Recupera los viajes realizados desde la estación Huron St & Franklin St y con duración menor a 500 segundos



```
> db.trips.find({ "start station name": "Huron St & Franklin St", tripduration: { $lt: 500 } })
< {
    _id: ObjectId('572bb8232b288919b68af1c4'),
    tripduration: 413,
    'start station id': 3116,
    'start station name': 'Huron St & Franklin St',
    'end station id': 3104,
    'end station name': 'Kent Ave & N 7 St',
    bikeid: 16757,
    usertype: 'Subscriber',
    'birth year': 1978,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.95826,
        40.73266
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -73.96150225,
        40.72057658
      ]
    },
    'start time': 2016-01-02T10:18:16.000Z,
    'stop time': 2016-01-02T10:25:09.000Z
  }
  {
```

5. Encontrar los 10 viajes más largos, ordenados de mayor a menor duración.

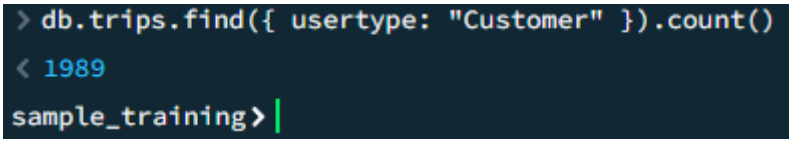


```
> db.trips.find().sort({ tripduration: -1 }).limit(10
< {
    _id: ObjectId('572bb8222b288919b68ac07c'),
    tripduration: 326222,
    'start station id': 391,
    'start station name': 'Clark St & Henry St',
    'end station id': 310,
    'end station name': 'State St & Smith St',
    bikeid: 18591,
    usertype: 'Subscriber',
    'birth year': 1979,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.99344559,
        40.69760127
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -73.98912867,
        40.68926942
      ]
    },
    'start time': 2016-01-01T00:58:20.000Z,
    'stop time': 2016-01-04T19:35:23.000Z
  }
  {
```

6. Encontrar viajes con estaciones que contengan Ave en su nombre.



7. ¿Cuántos viajes han realizado los usuarios de tipo Customer?.

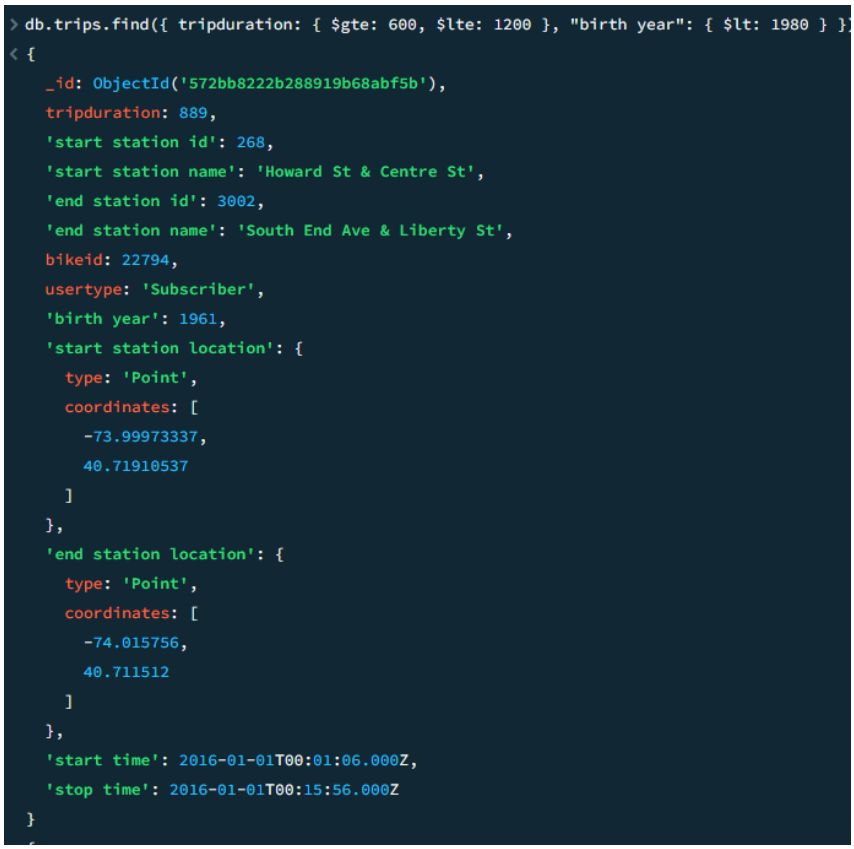


8. Recuperar los viajes realizados entre el 1 de enero y el 31 de marzo de 2016.



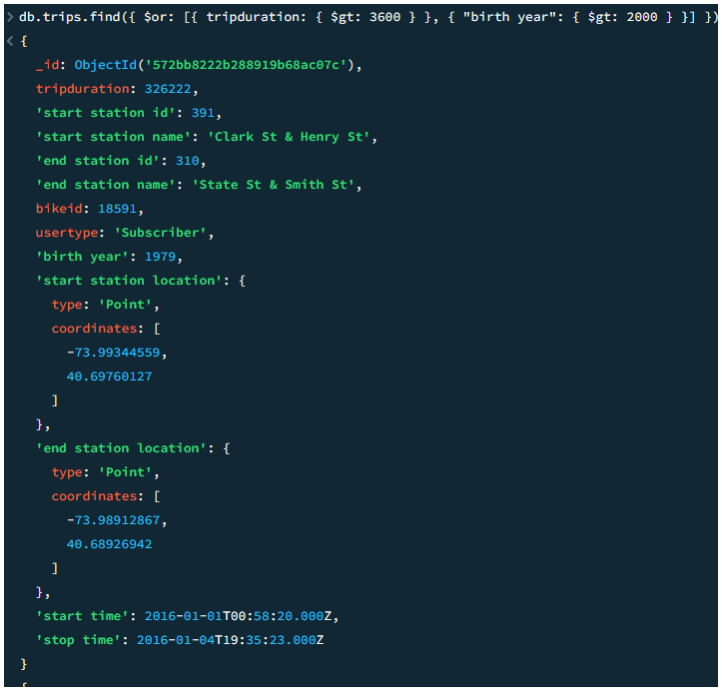
```
> db.trips.find({ "start time": { $gte: new Date("2016-01-01"), $lte: new Date("2016-03-31") } })
< {
    _id: ObjectId('572bb8222b288919b68abf5a'),
    tripduration: 379,
    'start station id': 476,
    'start station name': 'E 31 St & 3 Ave',
    'end station id': 498,
    'end station name': 'Broadway & W 32 St',
    bikeid: 17827,
    usertype: 'Subscriber',
    'birth year': 1969,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.97966069,
        40.74394314
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -73.98808416,
        40.74854862
      ]
    },
    'start time': 2016-01-01T00:00:45.000Z,
    'stop time': 2016-01-01T00:07:04.000Z
  }
```

9. Encontrar viajes con duración entre 10 y 20 minutos, hechos por usuarios nacidos antes de 1980.



```
> db.trips.find({ tripduration: { $gte: 600, $lte: 1200 }, "birth year": { $lt: 1980 } })
< {
    _id: ObjectId('572bb8222b288919b68abf5b'),
    tripduration: 889,
    'start station id': 268,
    'start station name': 'Howard St & Centre St',
    'end station id': 3002,
    'end station name': 'South End Ave & Liberty St',
    bikeid: 22794,
    usertype: 'Subscriber',
    'birth year': 1961,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.99973337,
        40.71910537
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -74.015756,
        40.711512
      ]
    },
    'start time': 2016-01-01T00:01:06.000Z,
    'stop time': 2016-01-01T00:15:56.000Z
  }
```

10. Encontrar los viajes que duraron más de una hora o que fueron realizados por usuarios nacidos después del año 2000.



```
> db.trips.find({ $or: [{ tripduration: { $gt: 3600 } }, { "birth year": { $gt: 2000 } }] })
< {
    _id: ObjectId('572bb8222b288919b68ac07c'),
    tripduration: 326222,
    'start station id': 391,
    'start station name': 'Clark St & Henry St',
    'end station id': 310,
    'end station name': 'State St & Smith St',
    bikeid: 18591,
    usertype: 'Subscriber',
    'birth year': 1979,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.99344559,
        40.69760127
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -73.98912867,
        40.68926942
      ]
    },
    'start time': 2016-01-01T00:58:20.000Z,
    'stop time': 2016-01-04T19:35:23.000Z
  }
```

11. Mostrar los 5 viajes más recientes, pero saltando los primeros 10 resultados.



```
> db.trips.find().sort({ "start time": -1 }).skip(10).limit(5)
< {
    _id: ObjectId('572bb8232b288919b68b1c29'),
    tripduration: 700,
    'start station id': 545,
    'start station name': 'E 23 St & 1 Ave',
    'end station id': 229,
    'end station name': 'Great Jones St',
    bikeid: 23922,
    usertype: 'Subscriber',
    'birth year': 1987,
    'start station location': {
      type: 'Point',
      coordinates: [
        -73.97809472,
        40.736502
      ]
    },
    'end station location': {
      type: 'Point',
      coordinates: [
        -73.99379025,
        40.72743423
      ]
    },
    'start time': 2016-01-02T19:30:34.000Z,
    'stop time': 2016-01-02T19:42:15.000Z
  }
```